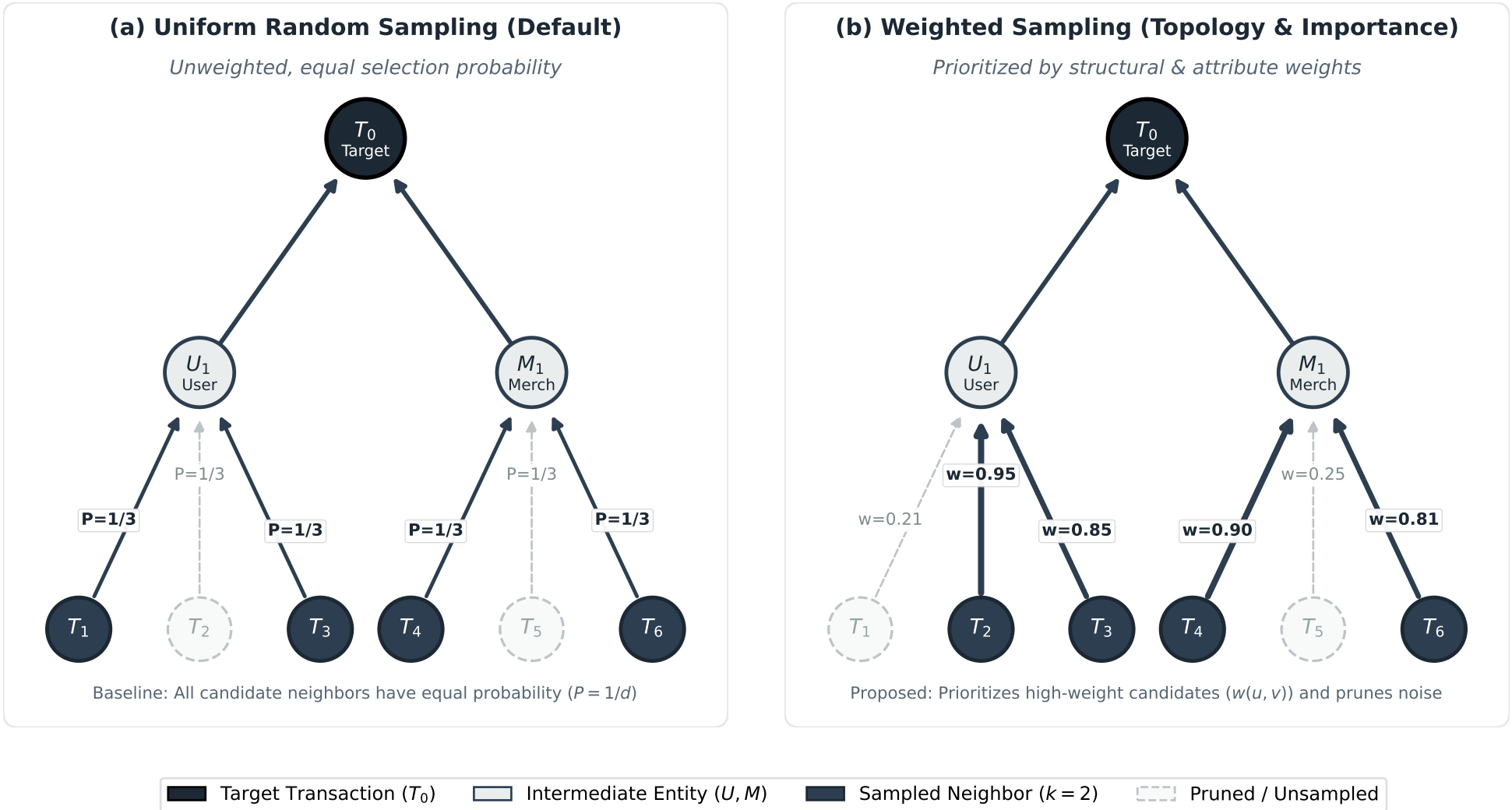
(a) Uniform Random Sampling (Default)
Unweighted, equal selection probability
T_0 Target
U_1 User
M_1 Merch
P=1/3
P=1/3
P=1/3
P=1/3
P=1/3
T_1
T_2
T_3
T_4
T_5
T_6
Baseline: All candidate neighbors have equal probability ($P = 1/d$)
(b) Weighted Sampling (Topology & Importance)
Prioritized by structural & attribute weights
T_0 Target
U_1 User
M_1 Merch
w=0.21
w=0.95
w=0.85
w=0.90
w=0.25
w=0.81
T_1
T_2
T_3
T_4
T_5
T_6
Proposed: Prioritizes high-weight candidates ($w(u, v)$) and prunes noise
Target Transaction (T_0)
Intermediate Entity (U, M)
Sampled Neighbor ($k = 2$)
Pruned / Unsampled